

Git Hands-On-3

Part A: Branching

1. Create a new branch “GitNewBranch”

```
$ cd GitDemo
$ git branch GitNewBranch
```

2. List all local and remote branches

```
$ git branch
```

Output:

```
GitNewBranch
* master
```

```
$ git branch -a
```

Output:

```
GitNewBranch
* master
remotes/origin/master
```

3. Switch to the newly created branch and add files

```
$ git checkout GitNewBranch
```

Output:

```
Switched to branch 'GitNewBranch'
```

```
$ git branch
```

Output:

```
* GitNewBranch
master
```

```
$ echo "This file was added on GitNewBranch" > branch_feature.txt
```

To confirm the file was created, I ran:

```
$ ls -la
```

Output:

```
total 12
drwxr-xr-x 1 Student 197121  0 Jul 25 12:00 .
drwxr-xr-x 1 Student 197121  0 Jul 25 10:02 ..
drwxr-xr-x 1 Student 197121  0 Jul 25 10:02 .git
-rw-r--r-- 1 Student 197121 38 Jul 25 12:00 branch_feature.txt
-rw-r--r-- 1 Student 197121 29 Jul 25 10:05 welcome.txt
```

4. Commit the changes to the branch

```
$ git add branch_feature.txt
$ git commit -m "Added branch_feature.txt on GitNewBranch"
```

Output:

```
[GitNewBranch 9d1c3aa] Added branch_feature.txt on GitNewBranch
1 file changed, 1 insertion(+)
create mode 100644 branch_feature.txt
```

5. Check the status with git status

```
$ git status
```

Output:

```
On branch GitNewBranch
nothing to commit, working tree clean
```

Part B: Merging

1. Switch to the master branch

```
$ git checkout master
```

Output:

```
Switched to branch 'master'
```

2. List differences between trunk and branch (CLI)

```
$ git diff master GitNewBranch
```

Output:

```
diff --git a/branch_feature.txt b/branch_feature.txt
new file mode 100644
index 0000000..2c4a1de
--- /dev/null
+++ b/branch_feature.txt
@@ -0,0 +1 @@
+This file was added on GitNewBranch
```

3. List visual differences using P4Merge

```
$ git config --global diff.tool p4merge
$ git config --global difftool.p4merge.cmd "p4merge \"$LOCAL\" \"$REMOTE\""
$ git difftool master GitNewBranch
```

Output:

P4Merge opened in a side-by-side visual diff window, with master's version (empty / file absent) shown on the left pane and GitNewBranch's version of branch_feature.txt shown on the right pane, with the added line highlighted in green. After reviewing the difference, I closed the P4Merge window to return to Git Bash.

4. Merge the source branch into the trunk

```
$ git merge GitNewBranch
```

Output:

```
Updating 3f2b9a1..9d1c3aa
Fast-forward
 branch_feature.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 branch_feature.txt
```

5. Observe the log after merging

```
$ git log --oneline --graph --decorate
```

Output:

```
* 9d1c3aa (HEAD -> master, GitNewBranch) Added branch_feature.txt on GitNewBranch
* 3f2b9a1 Added welcome.txt file
```

6. Delete the branch and observe git status

```
$ git branch -d GitNewBranch
```

Output:

```
Deleted branch GitNewBranch (was 9d1c3aa).
```

To confirm the branch was removed and check the final repository status, I ran:

```
$ git branch
$ git status
```

Output:

```
* master

On branch master
nothing to commit, working tree clean

$ git push origin master
```

Output:

```
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 335 bytes | 335.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0)
To https://gitlab.com/studentname/GitDemo.git
 3f2b9a1..9d1c3aa master -> master
```